transmit control signals from a ground operator. The ground control transmitter sent commands via radio waves, received by the onboard radio receiver and antenna, which then activated the mechanical control linkages connected to the rudder, elevator, and ailerons.

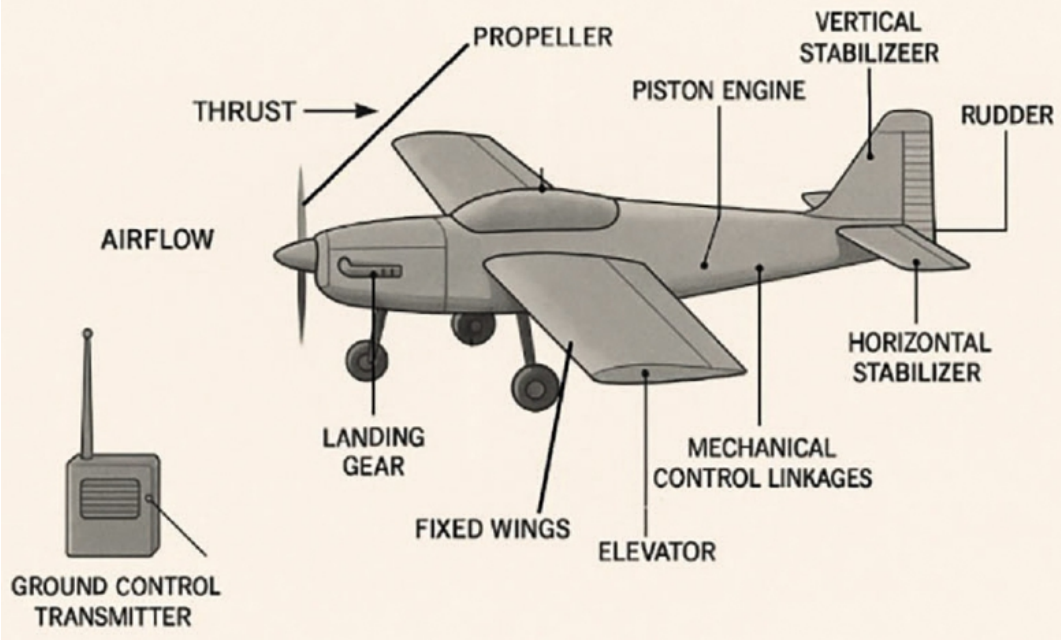


Figure 2.2 Principles and components of a first-generation drone.⏎

The fuselage housed the fuel tank, radio receiver, and engine, which powered the propeller to generate forward thrust. Lift was produced as air flowed over the fixed wings, while the horizontal and vertical stabilizers at the tail provided aerodynamic balance and directional control. The landing gear enabled take-off and landing on simple runways or open terrain.

Arrows in the figure indicate the directions of thrust and airflow, emphasizing the aerodynamic forces acting on the aircraft during flight. The diagram clearly demonstrates how early drones functioned through a manual, line-of-sight operation, where stability and navigation depended entirely on the operator’s input and visual observation.